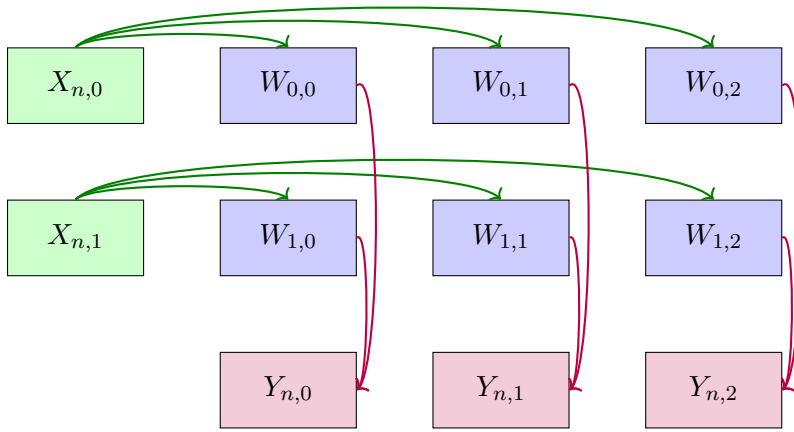


Introduction to Hardware Design

Arrays: Figures and Code Snippets

Profs. Sundeep Rangan, Siddharth Garg

0.1 array

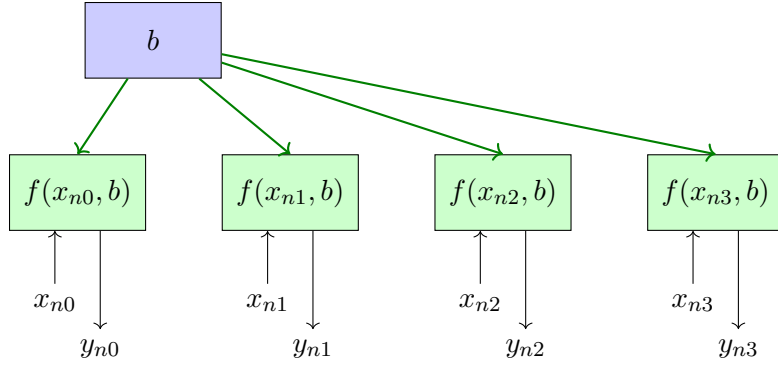


0.2 Fanout

Consider simple Vitis code:

```
for (int i = 0; i < N; i++) {  
    #pragma HLS PIPELINE II=1  
    for (int j = 0; j < M; j++) {  
        #pragma HLS UNROLL  
        y[i][j] = f(x[i][j], b);  
    }  
}
```

This can be visualized as:



```

bcopy[0] = b;
for (int j=1; j < M; j++) {
    bcopy[j] = bcopy[j-1];
}

for (int i = 0; i < N; i++) {
    #pragma HLS PIPELINE II=1
    for (int j = 0; j < M; j++) {
        #pragma HLS UNROLL
        y[i][j] = f(x[i][j], bcopy[j]);
    }
}

```

